

VIVEK BHARGAVA

Quantitative Researcher · Time-Series Forecasting, Production ML & Statistical Modelling
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PROFILE

Quantitative researcher in the **Mathematical Finance department of the Fraunhofer Institute for Industrial Mathematics**, focused on production ML, statistical modelling, and decision systems for industrial and financial clients. Builds and owns statistical/ML models end to end — from exploratory analysis and feature engineering to model design, validation, backtesting, deployment, and maintained software. Applied experience spans neural and econometric forecasting, large-scale anomaly detection, survival modelling, risk engines, and multi-criteria optimization systems. Combines computational statistics, econometrics, and statistical learning with an IIT Bombay mechanical-engineering foundation in physical systems, measurement, control, thermodynamics, fluids, and operations research.

CORE COMPETENCIES

Time Series & Forecasting: multi-horizon forecasting with calendar-structure embeddings (weekday, public holidays, bridge days, seasonality) and meteorological covariates — the feature architecture of load and power forecasting; ARIMA/SARIMA, VAR/VECM, darts regression forecasters, VARIMA; nested hierarchical clustering for stable high-dimensional grouping; rolling & expanding-window backtesting; Diebold–Mariano forecast-accuracy testing; Kalman filter & state-space; MIDAS & mixed-frequency architectures

Statistical & Machine Learning: dense neural networks & embedding architectures, tree ensembles (Random Forest, Gradient Boosting, XGBoost), regularised regression (Ridge/LASSO/Elastic Net), SVMs, reservoir computing (Echo State Networks), anomaly-detection ensembles, survival analysis (Weibull, Cox PH, Random Survival Forests), hierarchical & spectral clustering, PCA; nested & time-series cross-validation, hyperparameter optimisation, feature selection

Optimization & Decision Modelling: multi-criteria / multi-objective optimisation, mean-variance & GMVP portfolio construction, covariance estimation & shrinkage, constrained numerical optimisation (Nelder–Mead, interior-point), scenario analysis; regulatory-capital and HGB/IFRS accounting outputs

Production Engineering: Python data/ML stack (pandas, NumPy, scikit-learn, statsmodels, darts, SciPy, TensorFlow/Keras); clean architecture (Pydantic v2, pytest), CI-enforced quality gates (ruff, mypy, import-boundary linting), GitHub Actions / GitLab CI; production R (Shiny, JDBC against PostgreSQL/HANA); SQL, Git, Docker (working), Streamlit, MATLAB, LaTeX

Econometrics & Inference: OLS, IV/2SLS, MLE, GMM, QMLE, M-estimation; Wald/LR/LM testing; bootstrap inference; Monte Carlo simulation; cointegration (Johansen ML, ADF), Granger causality, impulse response & FEVD; SVAR identification

Risk & Volatility: VaR/CVaR/Expected Shortfall (parametric, historical, filtered historical simulation) with coverage backtesting; GARCH family (EGARCH, GJR, IGARCH, FIGARCH) with QMLE; Sharpe/Sortino/Calmar, drawdowns

PROFESSIONAL EXPERIENCE

Fraunhofer Institute for Industrial Mathematics — Quantitative Researcher Nov 2022 – Present
Mathematical Finance Department · Kaiserslautern, Germany · applied ML and econometric research delivered as production software for insurance, banking and automotive clients

Short-Horizon Neural-Network Forecasting — *Fraunhofer project · Python (TensorFlow/Keras, pandas, scikit-learn)*

- Contributed to a production forecasting system using dense neural networks with embedding layers for calendar structure (month, weekday, public holidays and bridge days, hour) alongside seven meteorological covariates — the feature architecture characteristic of load and power forecasting.
- Benchmarked the neural models against LSTM, encoder–decoder LSTM and gradient-boosting (LightGBM) baselines; worked the full ML lifecycle — feature engineering, standardisation, training and multi-horizon out-of-sample evaluation at 24/72/120-hour horizons.

Budget 360 — End-to-End Time-Series Forecasting System — *automotive OEM · Python (darts, statsmodels, SciPy, pandas)*

- Built, as sole developer, an end-to-end forecasting system over ~3,000 cost time series (percentile-based volume filtering isolating the ~150 highest-impact), with univariate (darts regression forecasters) and multivariate (statsmodels VAR / darts VARIMA) models under data sparsity.
- Applied **nested hierarchical clustering** (correlation distance, cluster-size control; SciPy) to form stable VAR groupings and control estimation risk in high dimensions — grouping correlated series into coherent, adequately-sized blocks so multivariate models estimate on well-conditioned covariance structures rather than the full sparse cross-section.
- Implemented rolling and expanding-window backtesting with Diebold–Mariano forecast-accuracy tests to compare model behaviour under non-stationarity and incomplete histories and guide model selection; owned the workflow from data extraction to decision-ready forecast delivery.

Risk Engine Re-engineering — MATLAB → Python — *major Luxembourg bank*

- **Led** the re-engineering of a legacy MATLAB risk engine into a production Python library in daily use by the bank's risk team — 30+ risk/return metrics (VaR, CVaR, Expected Shortfall, Sharpe/Sortino/Calmar, drawdowns) at full numerical parity.
- Built a mixed-frequency time-series synchronisation layer (daily/monthly/quarterly) handling missing data, irregular reporting periods and calendar alignment; shipped with full automated tests, GitLab CI/CD and API documentation.

Strategic Asset Allocation — Multi-Criteria Optimization — *German life insurer*

- **Own** the R Shiny decision-support layer of a four-criteria strategic asset-allocation system (return, volatility, Solvency II ratio, turnover distance; based on the published weighted-Tchebycheff box algorithm, *OR Spectrum*) in operative use at a German life insurer — interactive Pareto exploration across 26 asset classes, plus the feeding data pipeline including regulatory-capital and HGB/IFRS accounting outputs.

Warranty-Claim Anomaly Detection at Scale — *automotive OEM*

- Designed and deployed a five-model anomaly-detection ensemble scoring ~5.7M claims aggregated from ~80M raw claim and position records; supervised models (Random Forest, Gradient Boosting/XGBoost, neural network) trained on ~250K labeled claims, fused with Isolation Forest and group-conditional Z-score detectors via an L2-regularised linear stacking meta-learner.
- Resolved temporal label decay by training supervised models on the densely audited window while scoring the full population; deployed as a fit-once/score-forever pipeline with distributional guardrails gating database writes (HANA).

Survival Modelling & Cashflow Projection — *automotive & insurance clients*

- Developed survival models on ~108K vehicle records (Weibull baseline extended with Random Survival Forests) and quantified the model's reliable forecasting horizon (~1,300 days, beyond which estimates flatten) to prevent overconfident extrapolation for warranty and service decisions.
- Co-developed a cashflow projection framework with a nonparametric repayment estimator handling left-truncation — joint paper in preparation (*Nonparametric Loan Repayment Modelling Under Left-Truncation*, Fraunhofer ITWM).

Engineering & Delivery Practices — *research-to-production*

- Sprint-based Agile delivery (GitLab, Jira); peer code review via merge requests; built and maintained CI/CD pipelines with automated unit and integration testing; supervised student research assistants — defining work packages, reviewing merge requests, merging approved contributions.

Prior Experience — *India*

2013 – 2019

- **PwC (Financial Advisory)**: cashflow models, scenario analysis and bid structuring for infrastructure PPP projects. **Housing.com**: built an internal algorithmic call-routing module and KPI framework that raised effective utilisation of the agent base. **Indian Oil**: petroleum logistics operations. Internships: Reckitt Benckiser, Markem-Imaje, Tata Motors.

INDEPENDENT INVESTMENT RESEARCH & PERSONAL PROJECTS

Investment Research Platform — Solo-Built Python System — *delivered through an AI-augmented workflow · ongoing*

- Four-layer clean architecture (Pydantic v2, zero-I/O domain layer enforced by import-boundary linting in CI); multi-source market-data layer; Streamlit operational UI; shipped as 70+ tickets across 7 milestones, 387 commits, 867-test regression suite. Delivery runs through an AI-augmented workflow — LLM agents implement specification-driven development under deterministic state control, human-reviewed PR-only merges and CI-enforced quality gates.

RESEARCH & ACADEMIC WORK

MSc Thesis — Reservoir Computing for Macroeconomic Forecasting with Mixed-Frequency Data — *University of Konstanz · MATLAB, from scratch*

- Designed and compared **three Echo State Network architectures** for mixed-frequency forecasting of quarterly US GDP growth; implemented the full engine from scratch (~25 MATLAB functions) — leaky-integrator state updates, spectral-radius normalisation, sparse reservoir generation, closed-form ridge (Tikhonov) read-out.
- Benchmarked against U-MIDAS and VAR implemented from first principles with information-criteria lag selection; hyperparameter optimisation via grid search plus Nelder–Mead and constrained numerical optimisation under time-series-aware k-fold cross-validation.
- **Cluster-based portfolio construction (seminar)**: hierarchical & K-means clustering on the S&P 500 (205 assets × 443 months) reducing covariance-estimation dimensionality; GMVP core, strict rolling 60-month out-of-sample backtest with transaction costs, benchmarked against 1/N.
- **Research assistant & tutor** — statistical-learning-theory group (Prof. L. Grigoryeva), University of Konstanz. **Invited talk** — “AI Against Healthcare Fraud”, Data Science for Sustainable Finance and Economics 2025, HTW Berlin.

EDUCATION

University of Konstanz — MSc Economics (Econometrics & Applied Economics) 2019 – 2022

Grade 1.7 · Begabtenstipendium 2021 (academic excellence scholarship)

Indian Institute of Management Calcutta — MBA, Finance & Quantitative Methods 2015 – 2017

Indian Institute of Technology Bombay — B.Tech Mechanical Engineering 2009 – 2013

ACHIEVEMENTS, LEADERSHIP & LANGUAGES

JEE 2009: top 2.1% of 384,977 candidates (national engineering entrance examination, India). **Leadership (IIT Bombay)**: elected Cultural Councillor three consecutive terms; Organisational Citation (top 2 of 350) and Roll of Honor. **Languages**: English (fluent, IELTS 8.0), German (basic, A2).